

A Self-Improving Multimodal Multi-Agent Framework with Curriculum-Grounded Hybrid RAG for Personalized Learning through Interactive Visualizations, Adaptive Assessment, Audio Review, and Intelligent Note Generation

Abstract - Learners often move between unrelated tools to view diagrams, test their understanding, review audio, and recover handwritten notes. LearnX connects these activities through four independently deployable subsystems: interactive visualization, adaptive assessment, multilingual audio review, and intelligent note recovery. The pipelines combine anatomy-aware image generation and critique, curriculum-grounded question generation, Sinhala handwriting enhancement and OCR, hybrid document retrieval, localization, and text-to-speech. On a fixed 2,000-example Sinhala test set, Stage-2 fine-tuning reduced word error rate from 41.48% to 32.60% and character error rate from 11.10% to 8.21%. A small warm-state pilot measured 2.516 s p50 for Sinhala speech recognition, 1.726 s for BGE-M3 embedding, 0.912 s for reranking, and 5.528 s for Kokoro-82M, compared with 13.334 s for an MMS-TTS baseline. For visualization, reviewed feedback can update memory and skills only after held-out validation; extending this mechanism to the other subsystems remains future work.

Index Terms—multimodal multi-agent systems, curriculum-grounded hybrid RAG, adaptive assessment, interactive visualization, multilingual audio review

I. INTRODUCTION

Biology learning moves repeatedly between diagrams, questions, spoken explanations, and handwritten notes. These activities usually sit in separate applications, which makes context difficult to carry forward and limits adaptation from earlier evidence. Reliability is equally important: a system should not change its behaviour from a single favourable interaction. Reviewed updates, held-out evaluation, and rollback are needed before feedback can safely alter later behaviour.

Embodied assistants have been explored for anatomy education [7], while low-rank adaptation [1], [2], promptable segmentation [3], diffusion transformers [4], and reflective agent workflows [5], [6] provide relevant building blocks. LearnX studies how these capabilities can serve four independently developed subsystems under a shared approach to context, retrieval, orchestration, and validation.

The contributions are a catalog-driven, LoRA-adapted visualization pipeline; curriculum-grounded adaptive assessment; evaluated super-resolution and Sinhala OCR; multilingual spoken question answering with Stage-2 Sinhala ASR and hybrid retrieval; and a validation-gated feedback lifecycle that

permits reviewed visualization updates to reach skills or long-term memory.

II. RELATED WORK

A. Diffusion-Based and Agentic Image Generation

Rectified-flow transformers such as FLUX.1 [4] make instruction-following text-to-image generation practical, but a single forward pass gives no guarantee of factual or pedagogical correctness. Reflexion [5] and Self-Refine [6] motivate critique-and-retry patterns; the visualization subsystem uses an independent dual critic rather than self-critique.

B. Vision-Language Grounding and Segmentation

SAM 2 [3] produces masks from point or box prompts without making semantic claims; Qwen2.5-VL [8] provides grounded visual reasoning and localization used by the interactive visualization subsystem.

C. Parameter-Efficient Fine-Tuning

LoRA [1] and QLoRA [2] reduce the cost of adapting large models. LearnX uses LoRA for anatomy specification and directly fine-tunes OCR and super-resolution for Sinhala handwriting, each evaluated against non-fine-tuned baselines.

D. Retrieval-Augmented Generation and Agent Memory

Sentence-BERT [9] and HNSW [10] support embedding-based retrieval. The visualization feedback lifecycle separates raw interaction evidence from reviewed memory and skill updates; this context model is not claimed as currently implemented in VoiceLearn AI.

E. Multilingual Speech and Hybrid Retrieval

Whisper [15] provides a strong multilingual speech-recognition baseline. For document-grounded answering, retrieval-augmented generation separates evidence retrieval from generation [16]. VoiceLearn combines BGE-M3 dense retrieval [17], BM25 [18], RRF [19], and BGE-reranker-v2-m3 [24] before grounded answer generation.

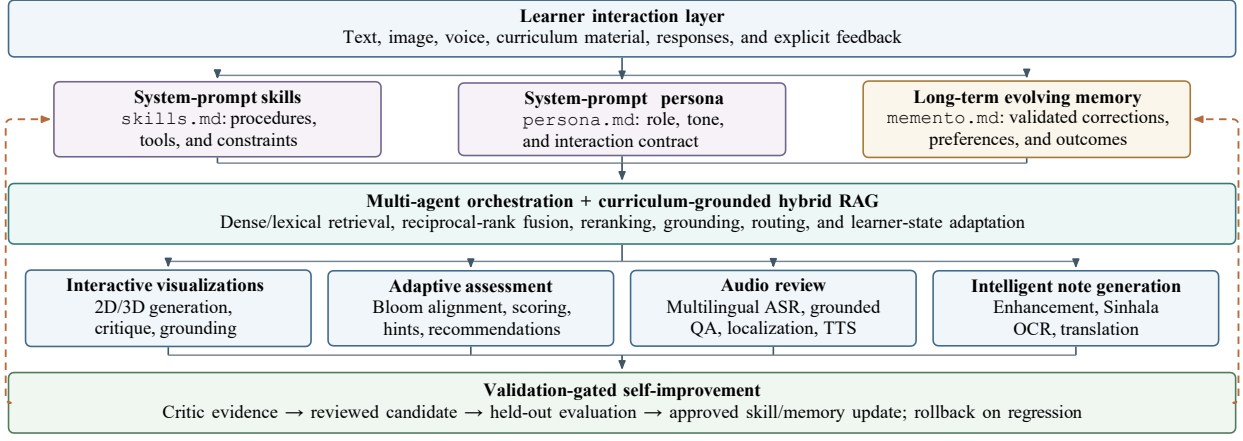


Fig. 1. LearnX architecture. Solid arrows show forward execution; dashed arrows show validation-approved updates to `skills.md` and `memento.md`. The feedback loop is currently implemented for visualization.

III. SYSTEM OVERVIEW AND SHARED DESIGN

LearnX is organized as a research monorepo rather than a single production application. Its four subsystems can be deployed independently, but they follow the same architectural vocabulary for context, orchestration, retrieval, and validation. Visualization, assessment, and audio review use stateful LangGraph workflows [11], with individual stages calling separately deployed services over FastAPI/HTTP. At runtime, controlled prompt instructions are combined with validated long-term memory. A shared learner profile and a single cross-component interface have not yet been implemented.

A. Agent Context and Feedback Lifecycle

Runtime context separates stable behaviour in `persona.md`, procedures and constraints in `skills.md`, and validated evidence in `memento.md`. Raw feedback may trigger a retry but cannot directly alter later behaviour. In visualization, reviewed corrections are deployed only after paired held-out evaluation, and the previous version remains available for rollback. The other subsystems do not yet implement this lifecycle.

B. Safety

Safety checks sit outside editable skill documents. Visualization uses a fail-closed deterministic pre-check plus contextual classification for borderline educational or medical requests.

IV. METHODOLOGY: THE FOUR SUBSYSTEMS

A. Interactive Biology Visualization

Internal codename EduVision. This subsystem turns a natural-language learning request into a validated, explorable visual artifact through five agents.

1) *Prompt Agent*: Qwen2.5-3B-Instruct [12], optionally paired with a LoRA adapter, maps a request to a general prompt or a structured *anatomy spec*. For five catalogued organs, the specification restricts organ identity, view, orientation, and required structures to canonical values. A deterministic builder then produces the final image prompt. The adapter uses a fixed

3,600/450/450 split generated from catalog contract v2.0.0 and is compared with the base model on the 450-example test set.

2) *Image Agent*: The validated prompt passes to FLUX.1-dev [4], a 12-billion-parameter rectified-flow diffusion transformer, run on tiered GPU classes (entry, mid, and high) to trade inference latency against cost.

3) *Evaluation and Reflection*: An independent evaluator reports CLIPScore [13] and a separate Qwen2.5-VL-7B-Instruct judgment [8]. The latter checks organ identity, view, orientation, required structures, and unwanted embedded text. Failed images are retried with the evaluator’s specific criticism.

4) *Interactive Agent*: After approval, the learner may tap or box-select a region. SAM 2 [3] produces a mask, while Qwen2.5-VL-7B-Instruct verifies the pixels, localizes the requested structure, and returns a brief explanation. Unsupported structures are omitted.

5) *3D Agent*: An approved 2D image can be converted into a textured, explorable 3D asset with Hunyuan3D-2 [14]: background-prepared shape generation followed by optional texture synthesis, with explicit GLB validation before return. The frontend renders the model with React Three Fiber and Three.js inside an Expo/React Native app, alongside SVG interactive labels for the 2D view.

B. Adaptive Assessment and Recommendation

Seven LangGraph agents personalize practice from uploaded PDF, DOCX, PPTX, and TXT files. They handle ingestion, evidence retrieval, grounded question generation, response scoring, difficulty adjustment, weak-topic recommendation, and analytics. Qwen2.5-7B-Instruct with a study-specific QLoRA adapter [30] generates MCQ, structured, and essay items aligned with Bloom’s taxonomy. The audited dataset [31] contains 1,500 master MCQs, paper-level splits of 1,182/90/76, and instruction-style splits of 5,910/450. Preparation normalizes options, audits answer keys and leakage, removes duplicates, and preserves paper-level separation. Three unsuccessful attempts receive progressively stronger Socratic hints; the fourth returns the answer with an explanation.

C. Intelligent Image-to-Notes Pipeline

For difficult Sinhala handwriting, fine-tuned Swin2SR [34] first enlarges each image by $4\times$, and fine-tuned TrOCR [35] transcribes it line by line. The Sinhala output is retained and translated into Tamil and English. Experimental visual-OCR and contextual-correction services are not in the active path. Evaluation compares the deployed pipeline with the non-fine-tuned *hasindu-k/sinhala-handwritten-notes-v3* baseline using CER, WER, exact match, and latency.

D. Interactive Audio Review

Internal codename VoiceLearn AI. This subsystem supports multilingual conversational revision from voice or typed questions, with answers grounded in learner-uploaded study documents. Typed input bypasses speech recognition and enters the same document-grounding path.

1) *Language-routed speech recognition*: Sinhala uses the Stage-2 fine-tuned Whisper-small Sinhala model based on Lingalingeswaran/whisper-small-sinhala [25]; Tamil uses the osmapi/tamil-asr-qwen3 mirror of the Tamil Qwen3 checkpoint originally published as lemuralabs/tamil-asr-qwen3 [26]; English/general fallback and code-mixed speech use Whisper Large V3 [15], pending dedicated code-switch evaluation. The Sinhala model is evaluated directly at the speech-recognition layer so WER/CER gains are not attributed to downstream correction.

2) *Query preparation*: Transcribed or typed questions are normalized while preserving language and intent. This stage is kept logically separate from speech recognition and retrieval.

3) *Hybrid retrieval and grounded answering*: Documents are parsed, chunked, embedded with BGE-M3 [17], and stored in ChromaDB. At query time, dense retrieval returns the top 10 candidates and BM25 [18] independently returns the top 10 lexical candidates. Rankings are fused with RRF [19], capped at 10, reranked by BGE-reranker-v2-m3 [24], and the top five evidence chunks are supplied to the base google/gemma-4-12B-it model [23]. No task-specific downstream answer-model fine-tuning is claimed.

4) *Localization and text-to-speech*: Qwen2.5-7B-Instruct [12] localizes grounded answers when required. English text-to-speech uses Kokoro-82M [20], Tamil uses IndicF5 [21], code-mixed output can use Indic Parler-TTS [28], and Sinhala uses SinhalaVITS-TTS-M2 [27]. Services are independently deployed behind FastAPI/Modal endpoints.

E. Cross-Cutting Infrastructure

Model-serving agents run as independent Modal functions behind FastAPI, allowing separate scaling, versioning, and GPU selection. Visualization stores curriculum and memory embeddings in PostgreSQL/Supabase with pgvector/HNSW [10]; assessment and audio review use ChromaDB. VoiceLearn fuses BGE-M3 and BM25 rankings with RRF before reranking and grounded generation.

TABLE I
EVALUATION METRICS BY SUBSYSTEM

Subsystem	Metrics already implemented
Visualization	Structure, relation, and orientation checks; clean-image compliance; label metrics; CLIPScore; VLM critic score; hard-failure rate; latency
Note recovery	CER; WER; exact-match accuracy; inference latency
Assessment	Grounding similarity/pass rate; hallucination rate; MCQ schema compliance; Bloom alignment; reserve trigger/recovery; latency
Audio review	Held-out Sinhala ASR WER/CER; warm p50/p95/RTF pilot; TTS speed baseline; localization chrF smoke test; formal retrieval, localization, MOS, and end-to-end metrics pending

V. EXPERIMENTAL SETUP

The following experiments use the datasets and evaluation procedures implemented in the project repository. Section VI includes only measurements produced by those procedures; planned evaluations are identified separately.

A. Datasets

The visualization dataset is generated from a catalog covering the brain, heart, kidneys, liver, and lungs. For each organ, the catalog records canonical structures, their relationships, permitted views, and supporting sources. Contract v2.0.0 fixes a deterministic 3,600/450/450 train/validation/test split. The adaptive-assessment experiments use the audited A/L Biology Adaptive Dataset created for this study [31], including both paper-level and instruction-style splits. Note recovery is tested on a held-out Sinhala handwriting set; the in-house enhancement and OCR checkpoints are anonymized for review [34], [35], while the public comparison checkpoint is cited separately [36]. Licensing, consent, and redistribution status for the custom handwriting collection must be resolved before public release. The Sinhala speech-recognition experiment uses SPEAK-ASR/openslr-sinhala-asr [32], derived from the OpenSLR Sinhala resource [33], with a fixed 16,000/2,000/2,000 split. Both Whisper checkpoints are evaluated on the same 2,000-example test partition using WER and CER.

B. Data and Model Availability

Repository identifiers for the adaptive dataset, adapter, and in-house note-generation checkpoints are withheld during double-blind review [31], [30], [34], [35]; the public Sinhala speech data and comparison checkpoint remain cited [32], [33], [36]. No verified public dataset repository was found for the generated visualization set or custom 623-sample handwriting collection; both remain unreleased pending licensing, consent, and redistribution review.

C. Evaluation Protocols

- 1) *Visualization ablation study*: six configurations independently toggle reflection, memory retrieval, skill rules, and dual-critic scoring, from a vanilla legacy pipeline to the full pipeline; the protocol runs 100 prompts with five seeds.
- 2) *Anatomy component study*: raw-base, agentic-pipeline, and fine-tuned-pipeline conditions are compared on the

same held-out prompts and seeds using structure recall, relation accuracy, orientation correctness, clean-image compliance, label metrics, hard-failure rate, and latency.

- 3) *Prompt LoRA paired evaluation*: base versus LoRA-adapted prompt agent on the frozen 450-example split (60-example pilot, then full set), with per-organ summaries, latency percentiles, hard-failure rates, and paired bootstrap confidence intervals.
- 4) *OCR evaluation*: fine-tuned Swin2SR+TrOCR versus the non-fine-tuned baseline on the held-out Sinhala set using CER, WER, exact match after Unicode NFC normalization, plus per-example latency.
- 5) *Sinhala ASR paired baseline evaluation*: non-Stage-2 versus Stage-2 Whisper-small on the same frozen 2,000-example test partition, with WER/CER after Unicode NFC normalization; the test is isolated from RAG, localization, and TTS.
- 6) *VoiceLearn serving pilot*: one warm-up is discarded, followed by three back-to-back measured calls; cold starts are excluded. Client wall-clock p50, nearest-rank p95, mean, sample standard deviation, and real-time factor are reported. With $n = 3$, p95 is the observed maximum and remains preliminary. Accelerator labels are configured Modal classes, not runtime-probed devices.
- 7) *Quality smoke tests*: one synthetic Sinhala utterance and one machine-drafted localization reference per tested language are diagnostic only.
- 8) *Retrieval validity gate*: Recall@5, MRR@10, and nDCG@10 are withheld because the current index contained 298 embeddings but only 18 distinct passages (duplication factor 16.6) and unrelated documents.

D. Baselines and Metrics

Visualization uses raw FLUX.1-dev and base Qwen2.5-3B-Instruct baselines. OCR uses *hasindu-k/sinhala-handwritten-notes-v3* under identical preprocessing. VoiceLearn’s main speech-recognition baseline is the pre-Stage-2 Sinhala Whisper-small checkpoint on the same held-out test set; pilot baselines include zero-shot Whisper Large V3 on one identical Sinhala utterance, MMS-TTS [22] on the same English synthesis text as Kokoro, and an untranslated copy-through localization floor.

VI. RESULTS

Results are reported by component so that each claim can be traced to its corresponding baseline and evaluation protocol.

A. Interactive Visualization: Quality and Latency

Table II first reports the component’s quality comparison against raw FLUX.1-dev on the same held-out prompts. VLM-judged axes are higher with smaller spread; CLIPScore is lower. The pipeline renders labels on every image versus none for the baseline. This remains a preliminary run pending confirmation of sample and seed counts.

Table III then reports warm-state per-agent latency for the same visualization component’s deployed Modal services. Image synthesis and gateway routing dominate latency; the

TABLE II
PIPELINE VERSUS RAW FLUX.1-DEV BASELINE

Metric	Raw ($\mu \pm \sigma$)	Pipeline ($\mu \pm \sigma$)	Δ	Rel.
Visual score	6.00 $\pm$ 3.46	8.67 $\pm$ 0.58	+2.67	+44.5%
Pedagogical	5.00 $\pm$ 3.46	7.67 $\pm$ 0.58	+2.67	+53.4%
VLM average	5.50 $\pm$ 3.46	8.17 $\pm$ 0.58	+2.67	+48.5%
CLIPScore	27.36 $\pm$ 1.67	23.71 $\pm$ 2.47	-3.65	-13.3%
Labels	0.0 $\pm$ 0.0	8.0 $\pm$ 0.0	+8.0	-

TABLE III
WARM-STATE LATENCY BY VISUALIZATION AGENT

Agent (Modal class)	Hardware	Latency
Prompt enhancement (PromptAgent)	CPU worker	30.2–31.5 s
Image synthesis (ImageAgent)	A10G GPU	82.0–95.0 s
Dual-critic evaluation (EvalAgent)	Standard worker	5.05–7.67 s
Interactive region (VLMAgent)	A10G GPU	8.60–8.97 s
API gateway (auto-labels)	Serverless mesh	38.9–62.0 s

TABLE IV
FINE-TUNED TROCR VERSUS BASE TROCR (227 HELD-OUT IMAGES)

Metric	Relative improvement
Character error rate (CER)	34.33% reduction
Word error rate (WER)	20.65% reduction
Inference speed	54.08% faster

TABLE V
WARM-STATE LATENCY BY TEXT-EXTRACTION AGENT

Agent/model	GPU	Latency (ms)
Fine-tuned TrOCR	T4	112
NLLB-200 translation	T4	210
Swin2SR 4 $\times$ enhancement	T4	240
Qwen2.5-7B validation	A10G	480
SinhaLM-4B context validation	A10G	580
Qwen2-VL-7B visual OCR	A10G	850

critic and interactive-region agents complete in under ten seconds.

B. Sinhala Handwriting OCR

The fine-tuned Swin2SR+TrOCR pipeline reports relative improvements over the non-fine-tuned baseline on 227 unseen images from a 623-sample Sinhala handwritten-OCR dataset. Figure 2(a) shows steady loss reduction and a minimum validation CER of 0.5085 at epoch 8, followed by a small increase at epoch 9; the deployed checkpoint therefore uses epoch 8.

C. Adaptive Assessment and Recommendation

A Modal-only automated benchmark compared four generation configurations on 50 cases (200 outputs) grounded in the NIE G.C.E. A/L Grade 12 Biology Resource Book. The run used Qwen2.5-7B-Instruct, the v4 adapter [30], a 0.55 grounding threshold, temperature 0, seed 26041, five warm-up calls per generation model, and an A10G class whose recorded runtime device was an NVIDIA A10. Figure 2(b) records the adapter’s minimum validation loss of 0.2600 at epoch 2.13;

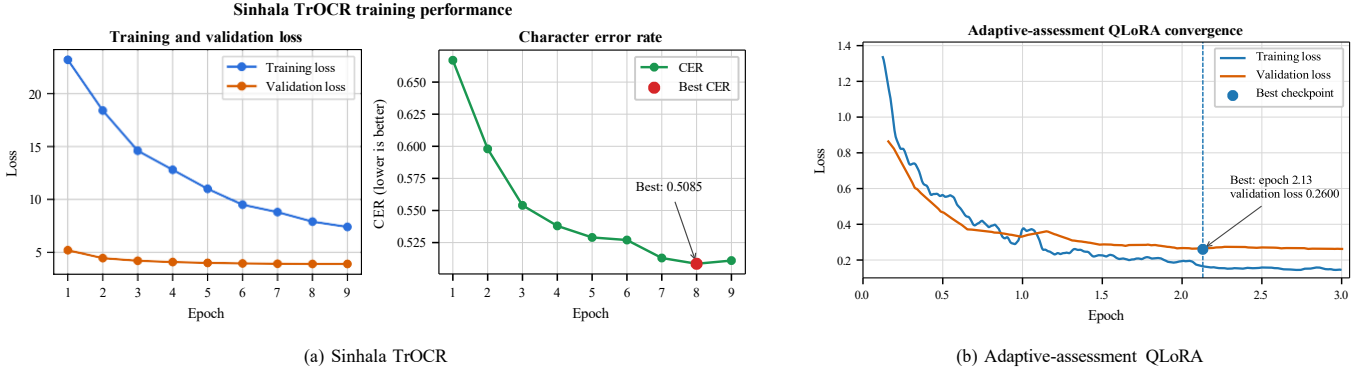


Fig. 2. Fine-tuning convergence reconstructed as vector graphics from the supplied training records. (a) TrOCR loss decreases across nine epochs, while held-out CER reaches 0.5085 at epoch 8. (b) Qwen2.5-7B-Instruct QLoRA reaches its minimum validation loss of 0.2600 at epoch 2.13.

TABLE VI
ADAPTIVE QUIZ AUTOMATED COMPARISON ($N = 50$)

Metric	ZS	RAG	Adapt.	Guard.
Grounding cosine ($\uparrow$)	0.5621	0.6080	0.6178	0.5993
Grounding pass ($\uparrow$)	60%	66%	76%	66%
Hallucination ($\downarrow$)	54%	24%	20%	22%
MCQ schema ($\uparrow$)	98%	94%	90%	94%
Bloom alignment ($\uparrow$)	36%	36%	32%	32%
Reserve replan trigger	-	-	-	52.0%
Reserve recovery	-	-	-	30.8%

ZS = zero-shot base; Adapt. = static adapter + RAG; Guard. = guarded adapter + RAG. Scaffolding recovery and learning gain were not measured because learner retries and a real pre/post study were unavailable.

TABLE VII
ADAPTIVE QUIZ ENDPOINT LATENCY EVIDENCE

Measure	Observed value
Visible successful calls	8/8
Startup	0 ms for all visible calls
Execution minimum	4.35 s
Execution median	4.71 s
Execution mean	5.01 s
Observed maximum	6.95 s
Accelerator	A10G configured; NVIDIA A10 recorded

Summary computed from the eight execution rows visible in the supplied Modal capture; it is deployment evidence, not a full latency distribution or an end-to-end learner study.

Table VI reports only automated component-level results, with arrows indicating the preferred direction.

D. VoiceLearn AI: Sinhala ASR

On the frozen 2,000-example test partition, Stage-2 reduced WER from 41.48% to 32.60% (8.88 percentage points; 21.41% relative) and CER from 11.10% to 8.21% (2.89 percentage points; 26.04% relative). These values quantify speech recognition only, not RAG, localization, or TTS quality.

E. VoiceLearn AI: Credit-Capped Deployment Pilot

One warm-up was discarded before three back-to-back calls. Table IX reports measured warm rows. Kokoro’s 5.528 s median was 58.5% below the 13.334 s MMS-TTS baseline, while the Qwen2.5 localizer dominated at 129.092 s p50.

TABLE VIII
VOICELEARN SINHALA ASR BASELINE VERSUS STAGE-2

Metric	Baseline	Stage-2	Abs. red.	Rel. red.
WER	41.48%	32.60%	8.88 pp	21.41%
CER	11.10%	8.21%	2.89 pp	26.04%

Sinhala VITS had a large p95 outlier. The $n = 1$ speech-recognition and localization checks remain diagnostic, and retrieval scores remain withheld after the corpus-validity audit. Publication-scale retrieval evaluation, answer-grounding evaluation, localization evaluation, TTS-naturalness evaluation, and end-to-end evaluation remain future work.

VII. DISCUSSION

The results do not favour a single configuration on every measure. The visualization pipeline scored higher on each VLM-judged axis, yet its CLIPScore was lower than the raw baseline. OCR improvements were observed on one held-out set of 227 images and should therefore be treated as component-level evidence. In adaptive assessment, static adapter+RAG produced the highest grounding similarity (0.6178), the highest pass rate (76%), and the lowest measured hallucination rate (20%). Those gains came with lower MCQ schema compliance (90%) and Bloom alignment (32%) than the zero-shot base (98% and 36%). The guarded configuration showed that reserve replanning and recovery can operate, although no conclusion about learning effectiveness can be drawn without learner data. For VoiceLearn, the paired 2,000-example Sinhala ASR experiment is the strongest result; the smaller serving pilot mainly reveals that localization is the current latency bottleneck.

VIII. LIMITATIONS AND THREATS TO VALIDITY

Several limitations constrain the conclusions. Automated critics provide useful diagnostic signals, but they do not replace expert or learner evaluation. The adaptive benchmark contains 50 cases drawn from one resource book, and learning gain cannot be estimated without real retries and pre/post testing. The available latency capture contains only eight visible calls, so it should not be interpreted as a stable production

TABLE IX
VOICELEARN AI WARM-STATE SERVING PILOT (3 SEPTEMBER 2026)

State	Model/service	Accel.	p50 (ms)	Max (ms)	Mean $\pm$ SD (ms)	RTF med.	RTF mean	Scope
ASR-English	Whisper Large V3 (faster-whisper, fp16)	T4	3,017	3,939	3,136 $\pm$ 751	0.40	0.42	Measured
ASR-Sinhala	Whisper-small Sinhala Stage-2 (fp16)	T4	2,516	3,369	2,709 $\pm$ 588	0.35	0.37	Measured
Retrieval embedder	BGE-M3, normalised 1024-D	T4	1,726	1,836	1,711 $\pm$ 134	-	-	Eight fixed texts
Retrieval reranker	BGE-reranker-v2-m3	T4	912	929	917 $\pm$ 10	-	-	Five candidates
Localization	Qwen2.5-7B-Instruct	T4	129,092	147,803	129,796 $\pm$ 17,666	-	-	Continuous warm [†]
TTS-English	Kokoro-82M	CPU	5,528	8,396	6,172 $\pm$ 1,982	0.73	0.82	Measured
TTS-Sinhala	SinhalaVITS-TTS-M2	T4	3,653	38,699	14,870 $\pm$ 20,649	0.51	2.09	One 38.7 s outlier [‡]
TTS baseline-English	MMS-TTS	T4	13,334	22,477	14,683 $\pm$ 7,214	1.87	2.06	Same English text

Note: Latencies are client-observed end-to-end wall-clock values and include TLS, request routing and JSON/audio transfer in addition to model compute; Table VIII separates the two where an endpoint exposes its own timer. One warm-up was discarded, then $n = 3$ measured calls were issued back-to-back; cold starts are excluded. At $n = 3$ the nearest-rank p95 coincides with the maximum observation and is reported as max. RTF is latency divided by audio duration, given in median and mean form. Accelerator classes are configured, not runtime-probed.

[†] The localization batch ran 389 s, exceeding the endpoint's 300 s idle scaled-down window. Because the calls were continuous the container was never reclaimed, so the batch remains warm-state; the window is an idle timeout, not a batch-duration limit.

[‡] The three Sinhala VITS calls were 2,257 / 3,653 / 38,699 ms. The mean, and hence the mean RTF of 2.09, is set by one call; the median RTF of 0.51 better represents steady-state behaviour.

distribution. Generated hints still need pedagogical review and explicit answer-leak checks. In addition, the adaptive dataset does not claim a blanket licence for every source-derived file; redistribution rights must be resolved before release [31]. Publication-scale evaluation is also still needed for Tamil, English, and code-mixed speech recognition.

IX. CONCLUSION AND FUTURE WORK

LearnX demonstrates how four independently deployable learning subsystems can share an architectural approach without overstating their present level of integration. Adapter+RAG improved grounding and reduced automated hallucination in the assessment benchmark, although schema compliance and Bloom alignment exposed clear trade-offs. The Sinhala ASR comparison provides evidence from a fixed 2,000-example test set, while the visualization, OCR, and serving results remain preliminary. The next stage of the work is to evaluate learning with real users, resolve dataset redistribution rights, and collect larger latency samples. The validation-gated update mechanism will also be extended beyond visualization only after component-specific held-out tests and rollback checks are in place.

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